

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&C) (Credit System) Degree Examinations

April 2023

21EC304 / 20EC304 – NETWORK THEORY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Using mesh analysis, find the current supplied by the battery for the circuit shown in Fig. Q1(a).

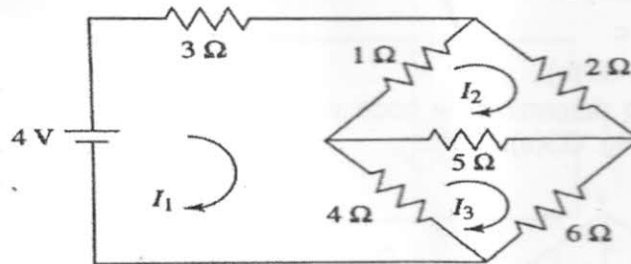


Fig. Q1(a)

10 L*3 1 1

- b) Reduce the circuit shown in Fig. Q1(b), to a single voltage source with a series resistance between the terminals A and B, by applying source transformation.

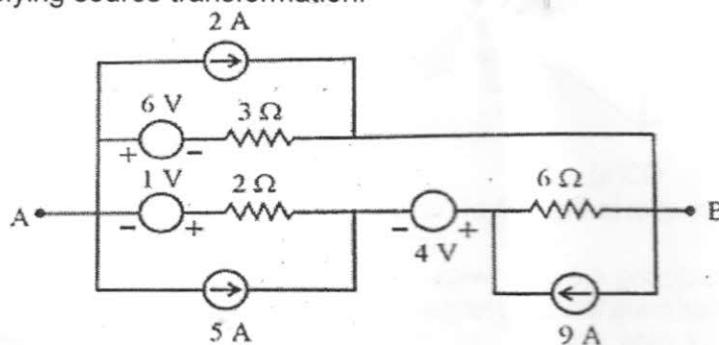


Fig. Q1(b)

10 L3 1 1

2. a) Find the equivalent resistance between A and B for the circuit shown in Fig. Q2(a) using star delta transformation.

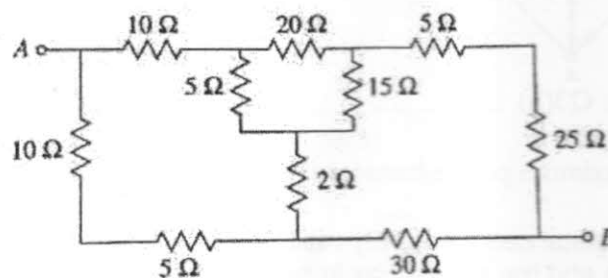


Fig. Q2(a)

10 L3 1 1

- b) For the network shown in Fig. Q2(b) draw the graph. Select tree with branches 2 and 4. Write cut-set matrix. Write equilibrium equation with node-pair voltage as variable. Solve these equations and find various branch voltages. The integers indicate branch numbers.

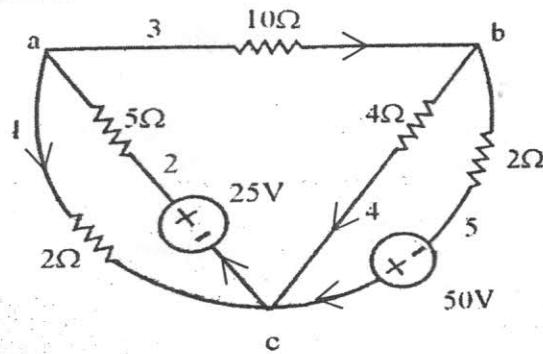


Fig. Q2(b)

10 L3 2 1

3. a) Apply node voltage analysis to determine the node voltages V_c and V_d for the circuit shown in Fig. Q3(a).

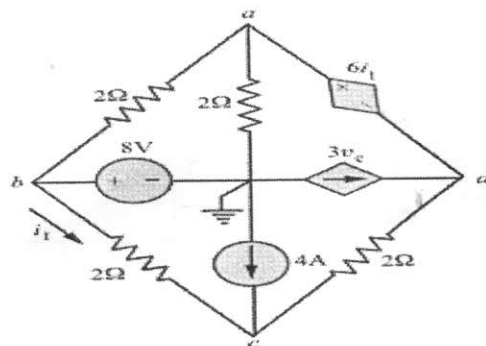


Fig. Q3(a).

10 L3 1 1

- b) The graph of the network is shown in Fig. Q3(b). write
 i. Incidence matrix
 ii. Tieset matrix by considering 4,5,6 as twigs.
 iii. Cutset matrix by considering 4,5,6 as twigs

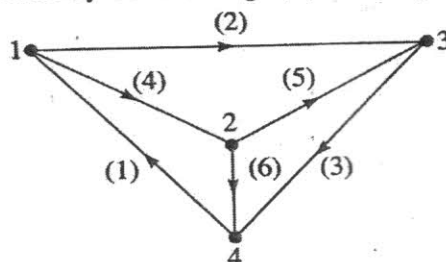


Fig. Q3(b)

10 L3 2 1

Unit - II

4. a) Determine the condition for maximum power transfer from source to load,
 i) When the load and source resistances are purely resistive.
 ii) When load is purely resistive and the source has impedance.

10 L2 3 1

- b) Determine the current through 10Ω resistor in the network shown in Fig. Q4(b)., using superposition theorem.

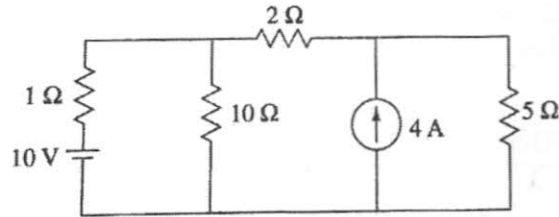


Fig. Q4(b)

10 L3 3 1

5. a) In the network shown in Fig. Q5(a). the switch is opened at $t=0$. Calculate the value of v , $\frac{dv}{dt}$ and $\frac{d^2v}{dt^2}$ at $t=0^+$

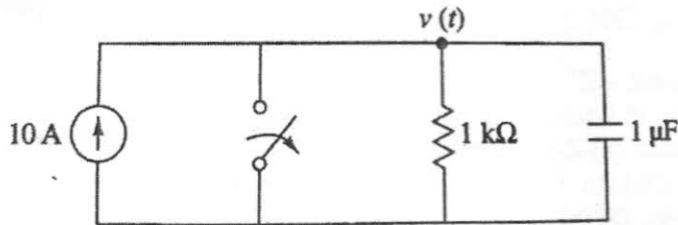


Fig. Q5(a)

10 L3 4 1

- b) For the network shown in Fig. Q5(b), Determine the current through 20Ω resistor by applying Thevenin's theorem.

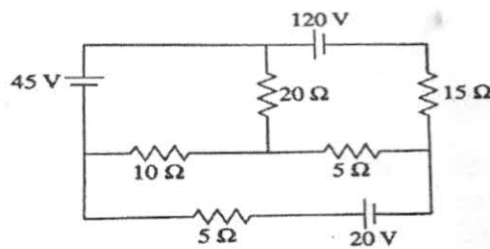


Fig. Q5(b)

10 L3 3 1

6. a) In the circuit shown in Fig. Q6(a), switch is closed at $t=0$. Find at $t=0^+$ the current through the circuit, its first and second order derivatives.

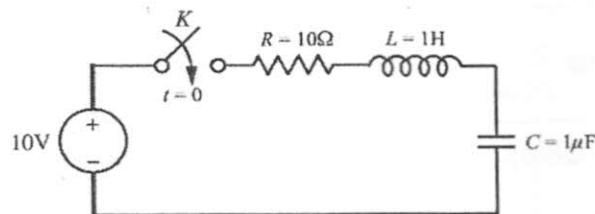


Fig. Q6(a)

10 L3 4 1

- b) In the circuit shown in Fig. Q6(b), the switch K is moved from position 1 to 2 at $t=0$. Steady-state condition having been reached at position 1. Solve for i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0^+$.

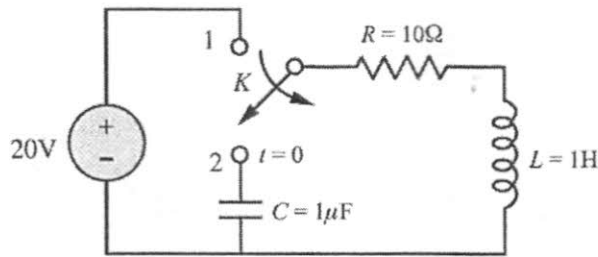


Fig. Q6(b)

10 L3 4 1

Unit – III

7. a) Define equations and draw the equivalent circuit of a two-port network for (i) Y parameters and (ii) Z parameters.
b) Define hybrid parameters. Obtain hybrid parameters for the two-port network shown in Fig. Q7(b).

10 L2 5 1

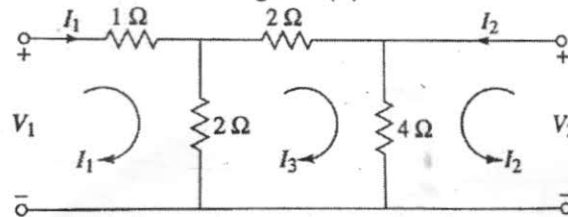


Fig. Q7(b)

10 L3 5 1

8. a) Define transmission parameters. Show that the transmission matrix of a cascade connection of two port network is the product of the transmission matrices of individual two-port networks.
b) Define Z parameters. Obtain Z parameters for the two-port network shown in Fig. Q8(b).

10 L2 5 1

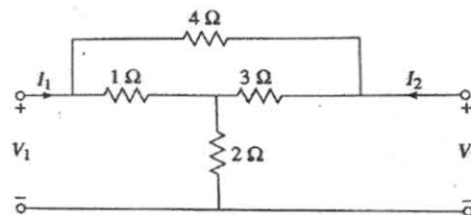


Fig. Q8(b)

10 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
